

Example 4.7

Compute the diffraction loss for the three cases shown in Figure 4.12. Assume $\lambda = 1/3$ m, $d_1 = 1$ km, $d_2 = 1$ km, and (a) $h = 25$ m, (b) $h = 0$, (c) $h = -25$ m. Compare your answers using values from Figure 4.14, as well as the approximate solution given by Equation (4.61.a)–(4.61.e). For each of these cases, identify the Fresnel zone within which the tip of the obstruction lies.

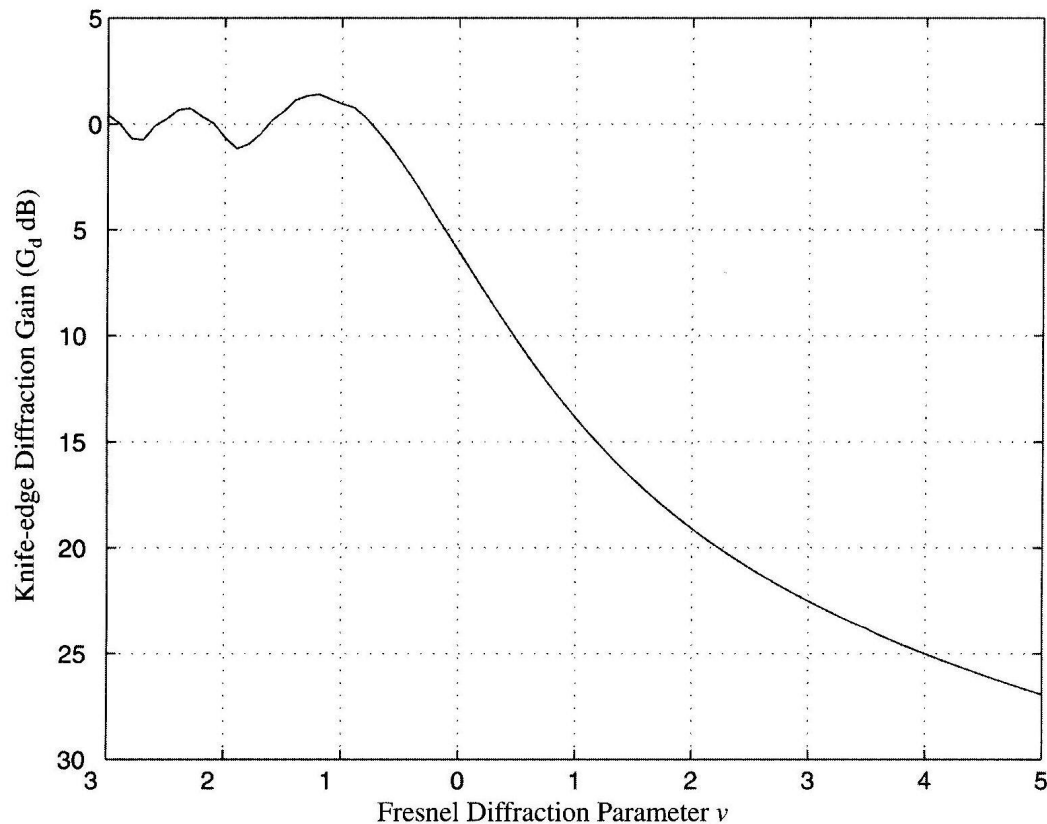


Figure 4.14 Knife-edge diffraction gain as a function of Fresnel diffraction parameter v .

Given:

$$\lambda = 1/3 \text{ m}$$

$$d_1 = 1 \text{ km}$$

$$d_2 = 1 \text{ km}$$

(a) $h = 25 \text{ m}$

Using Equation (4.56), the Fresnel diffraction parameter is obtained as

$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}} = 25 \sqrt{\frac{2(1000 + 1000)}{(1/3) \times 1000 \times 1000}} = 2.74.$$

From Figure 4.14, the diffraction loss is obtained as 22 dB.

Using the numerical approximation in Equation (4.61.e), the diffraction loss is equal to 21.7 dB.

The path length difference between the direct and diffracted rays is given by Equation (4.54) as

$$\Delta = \frac{h^2(d_1 + d_2)}{2 d_1 d_2} = \frac{25^2(1000 + 1000)}{2 \times 1000 \times 1000} = 0.625 \text{ m}.$$

To find the Fresnel zone in which the tip of the obstruction lies, we need to compute n which satisfies the relation $\Delta = n\lambda/2$. For $\lambda = 1/3 \text{ m}$, and $\Delta = 0.625 \text{ m}$, we obtain

$$n = \frac{2\Delta}{\lambda} = \frac{2 \times 0.625}{0.3333} = 3.75.$$

Therefore, the tip of the obstruction completely blocks the first three Fresnel zones.

(b) $h = 0$ m

Therefore, the Fresnel diffraction parameter $v = 0$.

From Figure 4.14, the diffraction loss is obtained as 6 dB.

Using the numerical approximation in Equation (4.61.b), the diffraction loss is equal to 6 dB.

For this case, since $h = 0$, we have $\Delta = 0$, and the tip of the obstruction lies in the middle of the first Fresnel zone.

(c) $h = -25$ m

Using Equation (4.56), the Fresnel diffraction parameter is obtained as -2.74 .

From Figure 4.14, the diffraction loss is approximately equal to 1 dB.

Using the numerical approximation in Equation (4.61.a), the diffraction loss is equal to 0 dB.

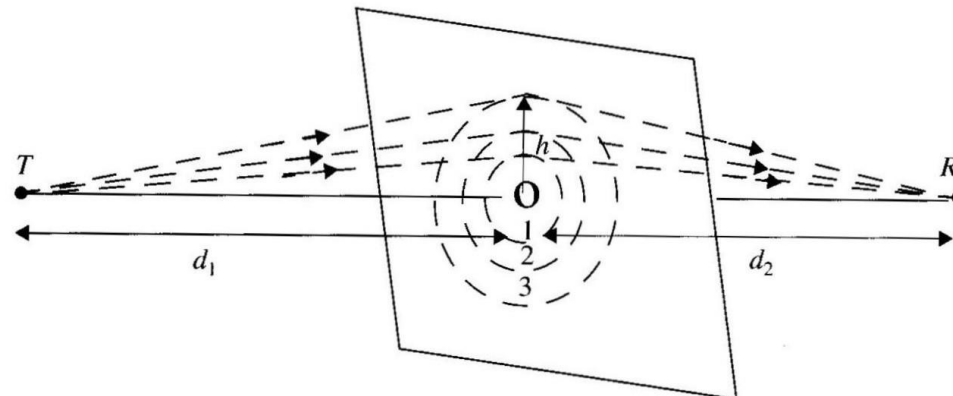
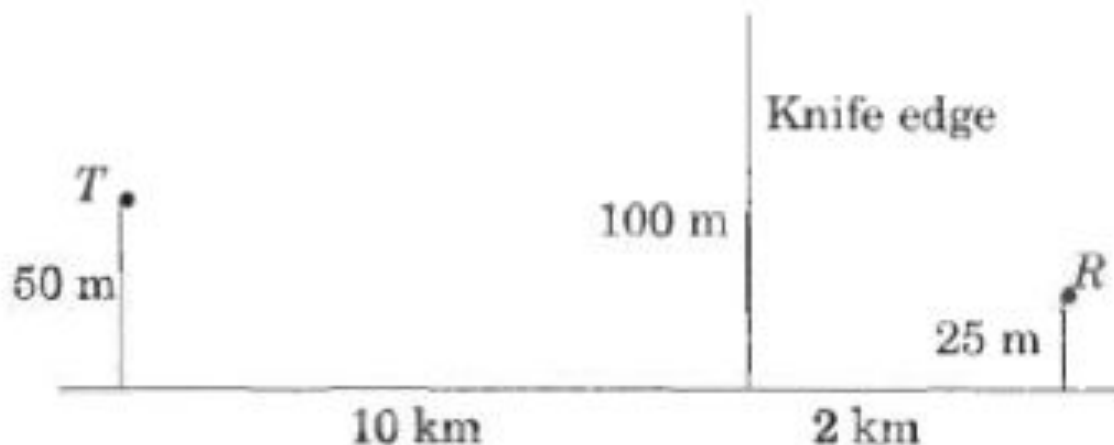


Figure 4.11 Concentric circles which define the boundaries of successive Fresnel zones.

Example 4.8

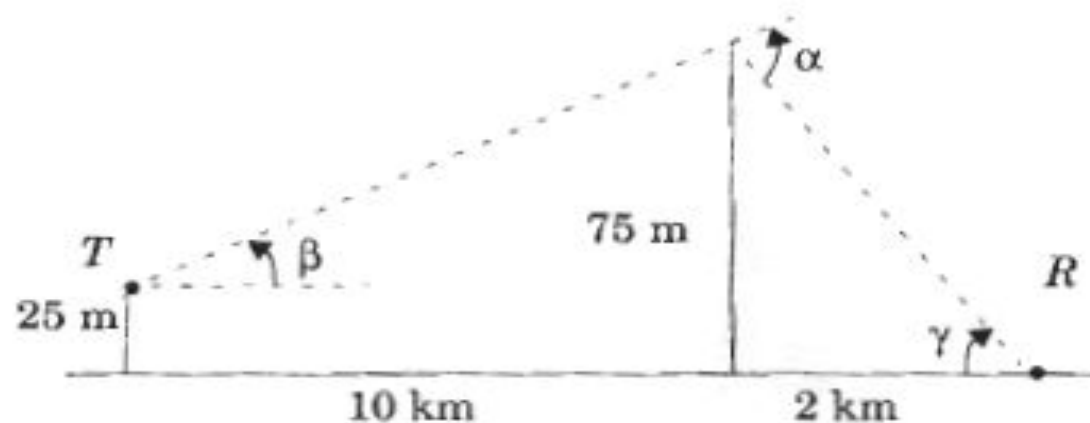
Given the following geometry, determine (a) the loss due to knife-edge diffraction, and (b) the height of the obstacle required to induce 6 dB diffraction loss. Assume $f = 900$ MHz.



Solution

(a) The wavelength $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{900 \times 10^6} = \frac{1}{3} \text{ m}$.

Redraw the geometry by subtracting the height of the smallest structure.



$$\beta = \tan^{-1}\left(\frac{75 - 25}{10000}\right) = 0.2865^\circ$$

$$\gamma = \tan^{-1}\left(\frac{75}{2000}\right) = 2.15^\circ$$

and

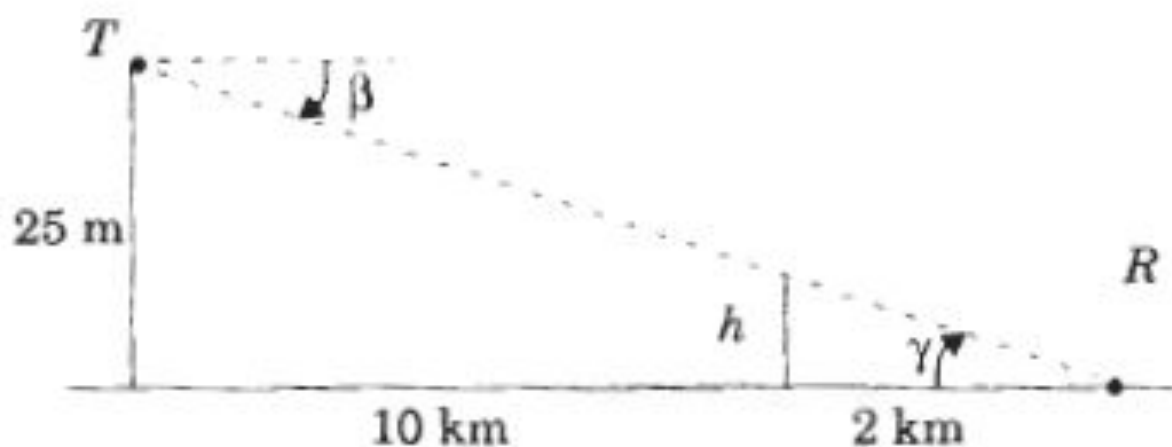
$$\alpha = \beta + \gamma = 2.434^\circ = 0.0424 \text{ rad}$$

Then using Equation (4.56)

$$v = 0.0424 \sqrt{\frac{2 \times 10000 \times 2000}{(1/3) \times (10000 + 2000)}} = 4.24.$$

From Figure 4.14 or (4.61.e), the diffraction loss is 25.5 dB.

- (b) For 6 dB diffraction loss, $v = 0$. The obstruction height h may be found using similar triangles ($\beta = \gamma$), as shown below.



It follows that $\frac{h}{2000} = \frac{25}{12000}$, thus $h = 4.16$ m.

Find the **median path loss** using Okumura's model for $d = 50\text{km}$ $h_{te} = 100\text{ m}$, $h_{re} = 10\text{ m}$ in a **sub-urban** environment. If the base station transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz, find the **power at the receiver** (assume a unity gain receiving antenna).

The free space path loss L_F can be calculated

$$L_F = 10\log\left[\frac{\lambda^2}{(4\pi)^2 d^2}\right] = 10\log\left[\frac{(3 \times 10^8 / 900 \times 10^6)^2}{(4\pi)^2 \times (50 \times 10^3)^2}\right] = 125.5\text{ dB}.$$

From the Okumura curves

$$A_{mu}(900\text{ MHz}(50\text{ km})) = 43\text{ dB}$$

and

$$G_{ARFA} = 9\text{ dB}.$$

$$G(h_{te}) = 20\log\left(\frac{h_{te}}{200}\right) = 20\log\left(\frac{100}{200}\right) = -6 \text{ dB.}$$

$$G(h_{re}) = 20\log\left(\frac{h_{re}}{3}\right) = 20\log\left(\frac{10}{3}\right) = 10.46 \text{ dB.}$$

The total mean path loss is

$$\begin{aligned} L_{50}(\text{dB}) &= L_F + A_{mu}(f, d) - G(h_{te}) - G(h_{re}) - G_{AREA} \\ &= 125.5 \text{ dB} + 43 \text{ dB} - (-6) \text{ dB} - 10.46 \text{ dB} - 9 \text{ dB} \\ &= 155.04 \text{ dB.} \end{aligned}$$

Therefore, the median received power is

$$\begin{aligned} P_r(d) &= EIRP(\text{dBm}) - L_{50}(\text{dB}) + G_r(\text{dB}) \\ &= 60 \text{ dBm} - 155.04 \text{ dB} + 0 \text{ dB} = -95.04 \text{ dBm.} \end{aligned}$$

The Hata Model

Standard formula for median path loss in urban areas is given by

$$L_{50}(\text{urban}) = 69.55 + 26.16 \log_{10} f - 13.82 \log_{10} h_1 - a(h_2) \\ + (44.9 - 6.55 \log_{10} h_1) \log_{10} d$$

For a small to medium sized city, the correction factor is given by

$$a(h_2) = (1.1 \log_{10} f - 0.7) h_2 - (1.56 \log_{10} f - 0.8) \text{ dB}$$

and for a large city, is give by

$$a(h_2) = 8.29(\log_{10} 1.54 h_2)^2 - 1.1 \text{ dB} \quad \text{for } f \leq 200 \text{ MHz} \\ = 3.2(\log_{10} 11.75 h_2)^2 - 4.97 \text{ dB} \quad \text{for } f \geq 400 \text{ MHz}$$

To obtain the path loss in decibels in a suburban are the formula is modified as

$$L_{50} = L_{50}(\text{urban}) - 2 [\log_{10}(f/28)]^2 - 5.4$$

And for the path loss in decibels in open areas the formula is modified as

$$L_{50} = L_{50}(\text{urban}) - 4.78 (\log_{10} f)^2 + 18.33 (\log_{10} f) - 40.98$$

Hata Model

- Empirical formulation of the graphical data in the Okamura model. Valid 150MHz to 1500MHz, Used for cellular systems
- The following classification was used by Hata:

■Urban area $L_{dB} = A + B \log d - E$

■Suburban area $L_{dB} = A + B \log d - C$

■Open area $L_{dB} = A + B \log d - D$

$$A = 69.55 + 26.16 \log f - 13.82 h_b$$

$$B = 44.9 - 6.55 \log h_b$$

$$C = 2(\log(f / 28))^2 + 5.4$$

$$D = 4.78 \log(f / 28)^2 + 18.33 \log f + 40.94$$

$$E = 3.2(\log(11.75 h_m))^2 - 4.97 \quad \text{for large cities, } f \geq 300 \text{MHz}$$

$$E = 8.29(\log(1.54 h_m))^2 - 1.1 \quad \text{for large cities, } f < 300 \text{MHz}$$

$$E = (1.11 \log f - 0.7) h_m - (1.56 \log f - 0.8) \quad \text{for medium to small cities}$$

Modified Cost 231

(European Cooperative for Scientific and Technical Research)

Mean Loss

P_t : transmitter power in watts

f : 1.5 – 2.0 GHz

h_1 : 30 to 200 m

h_2 : 1 to 10 m

d : 1 to 100 Km

$$L_{50}(\text{dB}) = 46.33 + 33.9 \log_{10} f - 13.82 \log_{10} h_1 - a(h_2) + (44.9 - 6.55 \log_{10} h_1) (\log_{10} d)^b + C_m$$

**Where C_m = 0 dB, for small/medium cities
= 3 dB, for large cities**

Field Strength

$$E \text{ in dB(dB mV/m)} = 63.1 + 10 \log_{10} P_t - 13.9 \log_{10} f + 13.82 \log_{10} h_1 + a(h_2) - (44.9 - 6.55 \log_{10} h_1) (\log_{10} d)^b - C_m$$

b as defined earlier

PCS Extension of Hata Model

- **COST-231 Hata Model, European standard**
- Higher frequencies: up to 2GHz
- Smaller cell sizes
- Lower antenna heights

$$L_{dB} = F + B \log d - E + G$$

$$F = 46.3 + 33.9 \log f - 13.82 \log h_b \quad f > 1500 \text{ MHz}$$

$$G = \begin{cases} 3 & \text{Metropolitan centers} \\ 0 & \text{Medium sized city and suburban areas} \end{cases}$$

Q.1 Consider a transmitter which radiates a sinusoidal carrier frequency of **1850 MHz**. For a vehicle moving **60 mph**, compute the received carrier frequency if the mobile is moving (a) directly towards the transmitter, (b) directly away from the transmitter, (c) in a direction which is perpendicular to the direction of arrival of the transmitted signal.

Solution: Given: Carrier frequency $f = 1850 \text{ MHz}$

Therefore, wavelength $= c/f_c = 0.162 \text{ m}$

Vehicle speed $v = 60 \text{ mph} = 26.82 \text{ m/s}$

(a) The vehicle is moving directly towards the transmitter.

The Doppler shift in this case is positive and the received frequency

is given by equation $f = f_c + f_d = 1850 \times 10^6 + \frac{26.82}{0.162} =$
 $= 1850.00016 \text{ MHz}$

(b) The vehicle is moving directly away from the transmitter. The Doppler shift in this case is negative and hence the received frequency is given by

$$f = f_c - f_d$$

$$f = 1850 * 10^6 - (26.82/0.162) = 1849.999834\text{MHz}$$

(c) The vehicle is moving perpendicular to the angle of arrival of the transmit-ted signal.

In this case, $\theta = 90^\circ$ $\cos\theta = 0$ and there is no Doppler shift.

The received signal frequency is the same as the transmitted frequency of 1850 MHz.